SIT SIH 2026
SMART INDIA HACKATHON 2026

- **Problem Statement ID -** SIH26217
- **Problem Statement Title -** Transforming AI Data Center Waste Heat into Sustainable Freshwater Generation
- **Theme -** Renewable/Sustainable Energy
- **PS Category-** Hardware
- **Team ID -** T103
- **Team Name -** AmritGen

AQUA COMPUTE

Generating Life, Sustaining Tomorrow

AmritGen
GENERATING LIFE, SUSTAINING TOMORROW

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❖ Proposed Solution (HDH Approach)

1. Waste Heat Extraction
AI servers & data centers generate high temperatures while computing. We **capture this waste heat** using a closed-loop cooling system and transfer it to the **HDH system**.

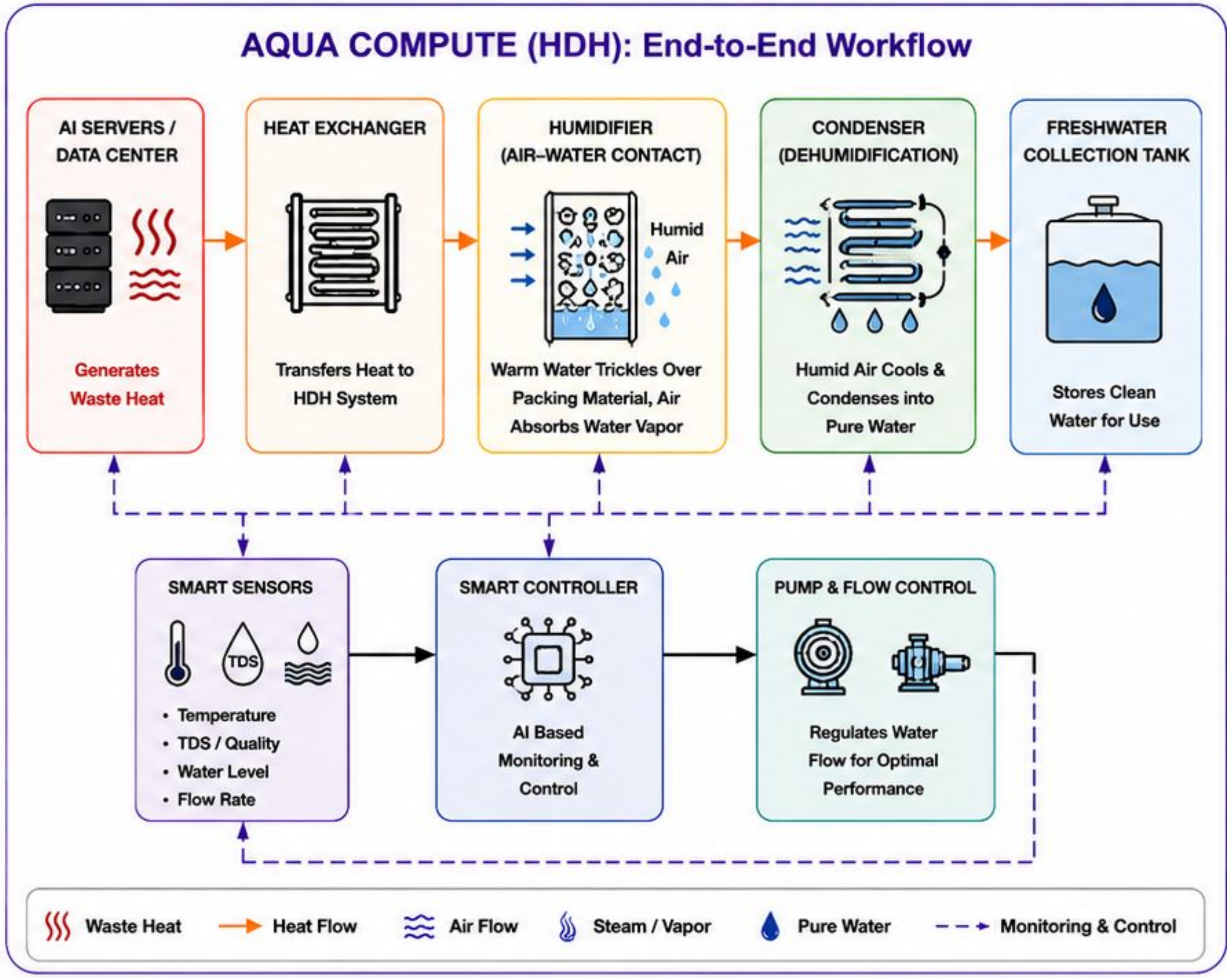
2. Humidification (Air–Water Contact)
Warm saline/brackish water is distributed over a packing material.
A fan pushes air through the humidifier where it absorbs water vapor.

3. Dehumidification (Condensation)
Humid air flows through the condenser.
The vapor cools down and condenses into **pure water**.

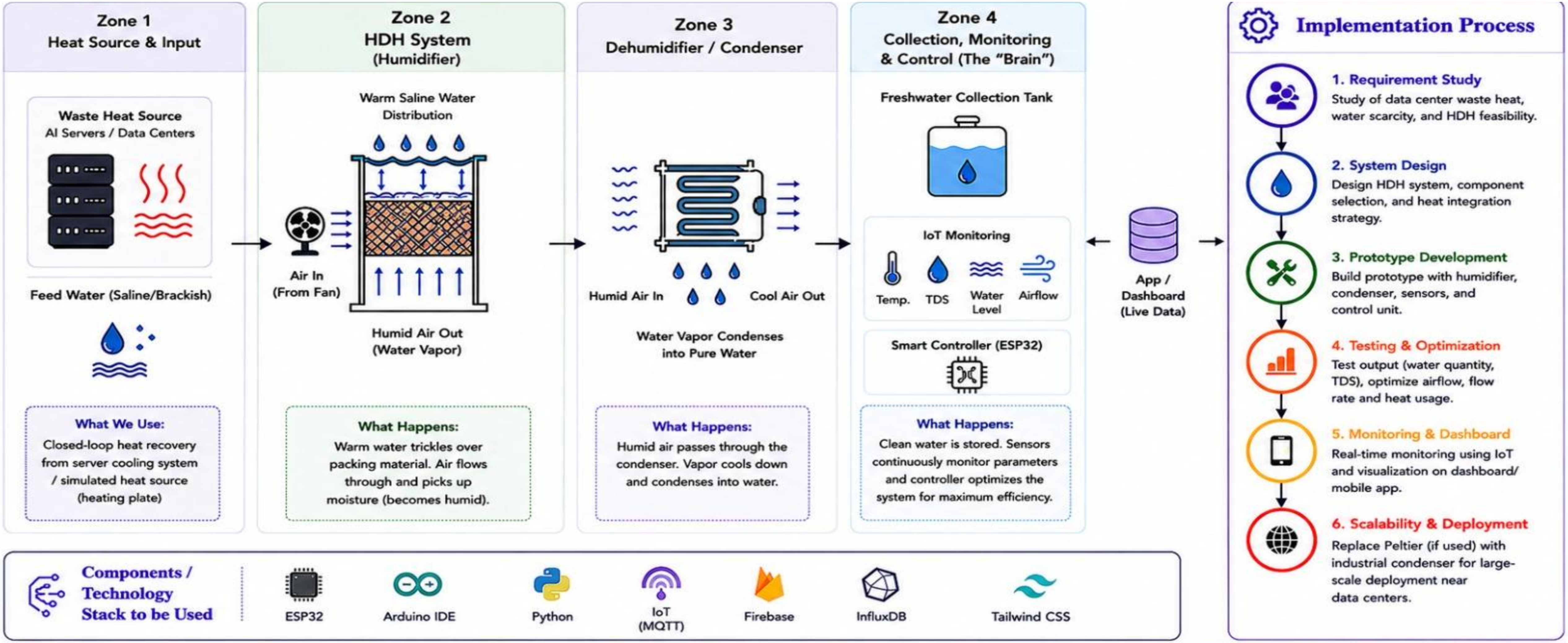
4. Collection & Storage
The condensed **freshwater** is collected and stored in a tank for safe use.

5. Smart Monitoring & Control
Sensors continuously monitor temperature, humidity, TDS and water level.
A smart controller optimizes airflow, water flow and heat utilization for **maximum efficiency** and safety.

6. Sustainable Impact
Converts waste heat into a valuable resource – **clean water**.
Low-cost, scalable and **eco-friendly** solution for water-scarce regions.



TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

Economic- Prototype cost around 2,500 – 3000 rs

Scalability- HDH can be scaled using larger packed humidifiers, air/water blowers, heat exchangers and modular condenser units

Challenges & Strategic Solutions

Potential challenges

1. 1) low-grade server waste heat may not provide sufficient temperature rise for high water production.
2. 2) Continuous evaporation can increase salinity and cause salt deposition, requiring flushing and periodic cleaning.

Strategies

1. 1) Using a thermal buffer to smooth variations in server waste heat.
2. 2) Using corrosion-resistant packing and monitoring salinity.
3. 3) Design the system as modular HDH units so capacity can be increas-ed without redesigning the complete plant.

IMPACT AND BENEFITS

Potential Impact on Targeted Audience

01 Water Security
Provides a reliable source of clean water for industries, communities, and water-scarce regions.

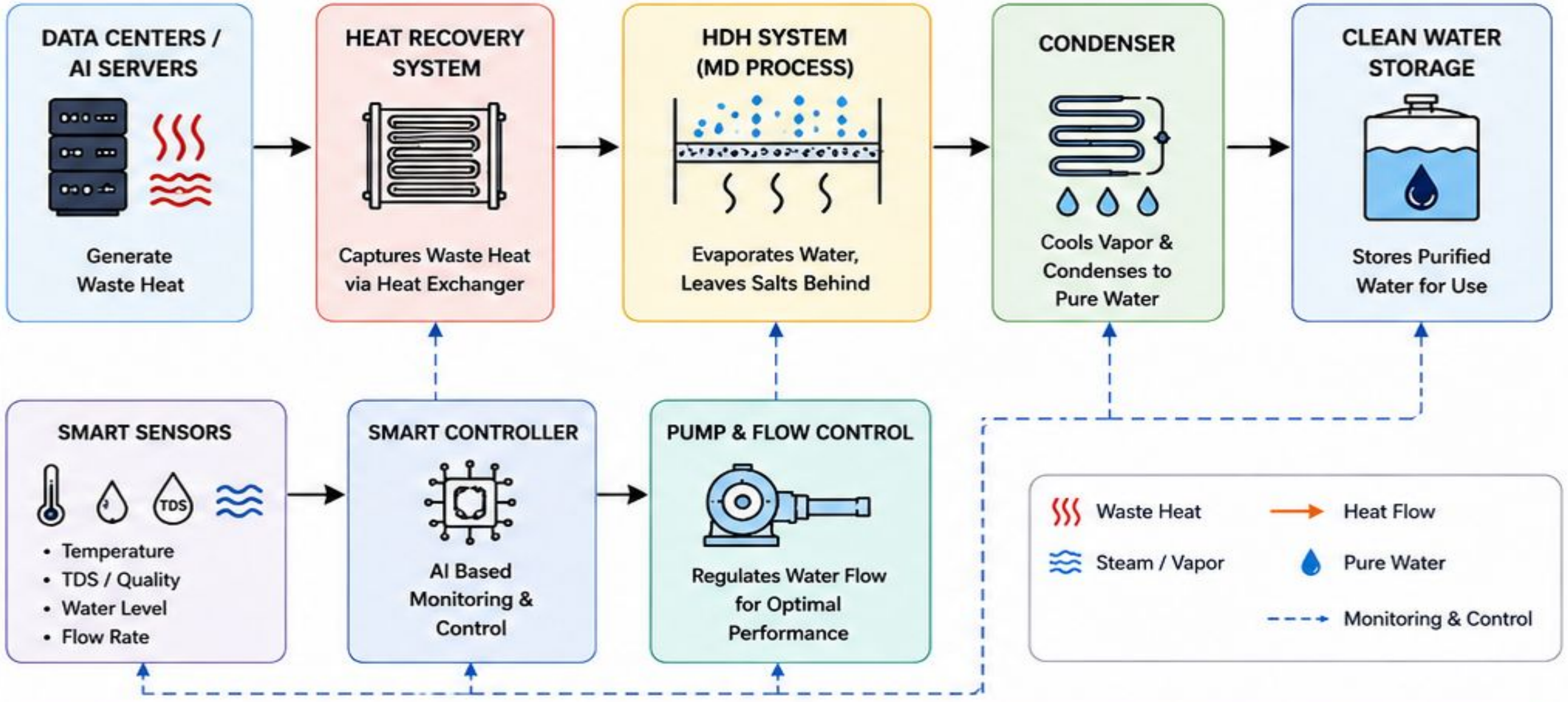
02 Environmental Impact
Reduces thermal pollution and conserves natural water resources through waste heat recovery.

03 Economic Benefits
Lowers operational costs by utilizing waste heat and reduces dependency on traditional desalination methods.

04 Social Impact
Improves access to clean water, enhancing public health and quality of life in underserved communities.

05 Scalability & Adaptability
Modular design enables easy scaling and adaptation across industries, locations, and climate conditions.

Benefits of the Solution



Sustainable Water Production
Generates clean water using waste heat with minimal resources.

Eco-Friendly Solution
Reduces carbon footprint and supports sustainable practices.

Reliable & Safe Operation
Smart monitoring ensures consistent water quality and system safety.

Cost Effective & Efficient
Lower operating costs through energy recovery and automation.

Community Empowerment
Improves water availability and supports community well-being.

RESEARCH AND REFERENCES

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RESEARCH

- **Studied** MD, HDH and MED desalination technologies using waste heat.
- **Feasibility Analysis:** Compared MD and HDH on cost, scalability, maintenance and efficiency; selected HDH.
- **Performance Benchmarking:** Analyzed water recovery, energy efficiency and heat utilization from research papers.
- **Prototype Validation:** Developing a small-scale HDH prototype to validate the concept.

REFERENCE

- Desalination Tower: An Alternative to Cooling Tower- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7216023
- A dual-purpose seawater evaporation system for data center cooling and freshwater production: design and performance analysis- https://eurekamag.com/research/101/055/101055378.php
- Sustainable thermodynamic innovations and energy-efficient hybrid designs in Humidification–Dehumidification (HDH) desalination: A comprehensive review- https://www.sciencedirect.com/science/article/abs/pii/S1364032126003734